COVER PAGE (paste into the template cover). Keep the template's PJDSC 2026 header and theme line.

BANTAY-BAHA

A Graph-Based Early Warning and Response System for Flood-Linked Disease Outbreaks

Trace the Pattern, Target the Cure: Advancing Public Health Through Data Science

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EXECUTIVE SUMMARY

246 words (limit 250)

Eight to nine tropical cyclones cross the Philippines each year, and flood-linked disease follows within weeks: Typhoon Ondoy (2009) was followed by 2,089 leptospirosis cases and 162 deaths, and the 2019 dengue epidemic reached 146,062 cases by July. Response still begins after hospitals fill, and the communities most at risk are often the ones a storm has cut off from care.

Bantay-Baha ("flood watch") is a decision-support dashboard that tells DOH regional offices and LGU health officers, before or immediately after a typhoon, which provinces are most likely to see a disease surge and which of those will lose access to a hospital. It links three graphs: (1) a flood-exposure layer from satellite-observed Sentinel-1 flood extent, LiPAD hazard maps and river-routed rainfall; (2) a causal lead-lag outbreak graph learned from province-week DOH dengue surveillance with PCMCI+, which identifies sentinel provinces that rise first; and (3) a road-and-RoRo ferry network that simulates storm-closed routes to measure lost hospital access. The first two feed a quantile gradient-boosted forecast of 1- to 4-week-ahead incidence, benchmarked against time-series foundation models and scored as international forecast hubs do. Surge probability and the population cut off from care combine into one priority ranking with recommended action tiers.

We will backtest on Typhoons Ompong (2018), Tisoy (2019), Rolly (2020) and Ulysses (2020). No Philippine tool conditions outbreak risk on a specific storm and joins it with storm-specific loss of healthcare access; Bantay-Baha moves response from reacting to outbreaks to getting ahead of them.

KEYWORDS

early warning system, dengue, leptospirosis, flooding, typhoon, graph analytics, causal discovery, probabilistic forecasting, healthcare access, Philippines

BACKGROUND

294 words

About 20 tropical cyclones enter the Philippine Area of Responsibility each year and eight or nine cross the country, mostly between July and October (PAGASA, n.d.). Each storm brings flooding, and flooding brings disease. After Typhoon Ondoy in 2009, more than 2,089 people were treated for leptospirosis and 162 died, over five times the national leptospirosis deaths of 2008 (Balbuena et al., 2010); Manila hospital data show the leptospirosis risk from rainfall peaks two weeks later and is mediated by flooding (Matsushita et al., 2018). Dengue rises with the same rains: in 2019 the DOH declared a national epidemic after 146,062 cases and 622 deaths from January to July (The Philippine Star, 2019). Recent reviews call for a unified surveillance registry and tools that prioritise where to act (Interior et al., 2025; Labana, 2026).

The data needed to act early exist but are scattered across surveillance, hazard, climate and facility datasets. Dengue forecasting from climate has been studied repeatedly in the Philippines, from province-level early warning (Buczak et al., 2014) to Metro Manila machine learning (Carvajal et al., 2018) and a nationwide comparison of eight forecasters (Olana et al., 2025). Flood-driven loss of hospital access has been modelled for Metro Manila, where a 15 cm flood leaves about 22% of residents without access to higher-level care (Tariverdi et al., 2023), and dengue source regions have been identified by synchrony analysis (van Panhuis et al., 2015). These strands have never been joined: no tool conditions surge risk on a specific typhoon, models nationwide loss of access including ferry links, or combines the two into one ranking for health officers. The problems share a structure that graph methods model directly: water flows along rivers, outbreaks move between connected places, and patients travel along roads and ferries.

THE PROBLEM

110 words

Philippine health authorities have no tool that anticipates, before or immediately after a typhoon, where flood-linked disease surges will occur and which affected communities will be cut off from care. Prophylaxis, vector control and medical teams are deployed after cases rise, and isolated communities are reached last. Existing dengue models are not storm-conditioned, give point forecasts without calibrated uncertainty and ignore whether patients can reach a hospital; hazard tools such as UP NOAH carry no health layer.

Guiding question: Can observed flood extent, a causal graph of outbreak spread and a storm-disrupted transport network, combined with calibrated machine-learning forecasts, rank the provinces that need help first, early enough to act?

OBJECTIVES

195 words. Lead-in sentence, then five SMART bullets.

By the PJDSC 2026 final-round submission, the team will:

- By 15 October 2026, assemble a province-by-week panel for all 81 provinces, 2016 to 2020, joining DOH-Epi dengue cases, IMERG rainfall, Sentinel-1 flood fraction, LiPAD hazard share and IBTrACS typhoon exposure on PSGC codes.
- By 20 October 2026, learn a lead-lag outbreak graph with PCMCI+, identify sentinel provinces (PageRank) and communities (Louvain), and test whether sentinel features reduce follower-province forecast error out of sample.
- By 20 October 2026, forecast 1- to 4-week-ahead dengue incidence per province with a quantile gradient-boosted ensemble, achieving a relative Weighted Interval Score below 1.0 against the seasonal-naive baseline and 90% interval coverage of 80 to 95% on rolling-origin evaluation over 2018 to 2019.
- By 20 October 2026, quantify storm-specific loss of hospital access for every city and municipality with a friction-surface and a road-plus-RoRo network method for Ompong, Tisoy, Rolly and Ulysses, with sensitivity bounds and the Metro Manila benchmark of Tariverdi et al. (2023).
- By 24 October 2026, deploy the dashboard that returns a ranked province list with calibrated risk, access loss, sentinel status, drivers and an action tier for any historical or shifted track in under one minute, and collect System Usability Scale scores from three target users.

SCOPE & LIMITATIONS

173 words

Scope. Nationwide, all 81 provinces at weekly resolution for 2016 to 2020, matching the DOH-Epi dengue file on HDX; access analysis per city and municipality, summarised per province as the share of population cut off. Dengue is the primary disease because consistent subnational data exist. Leptospirosis is a stretch goal: no open subnational series exists, so it enters as a national weekly lag analysis and, if DOH weekly surveillance reports (2022 to 2026) are digitised in time, as a region-week series. Storm scenarios use historical IBTrACS tracks and shifted variants. Users are DOH regional offices, LGU health offices and local DRRM offices.

Limitations. Surveillance data carry reporting delays and under-reporting, and COVID-19 collapsed 2020 reporting, so 2020 is used for access scenarios only. Sentinel-1 misses urban and vegetated floods and can miss the peak. OpenStreetMap completeness outside Luzon is uneven, so two access methods are reported. PCMCI+ links are statistical dependencies, not proof of transmission. The tool

supports decisions and does not replace clinical or field judgment; it uses only aggregated, non-identifiable data.

METHODOLOGY

645 words. Sub-headings are bold run-in text in the template.

Bantay-Baha builds three graphs, turns each into features for one province-week table, and combines a calibrated disease forecast with storm-driven access loss into one priority ranking (Figure 1, Appendix A; data sources in Table 1, Appendix B).

Layer 1: Flood exposure (prevent)

For each typhoon, Sentinel-1 radar before-and-after change detection in Google Earth Engine (UN-SPIDER recommended practice) yields the flooded fraction and flooded population (WorldPop 100 m) per province and municipality, cross-checked against Copernicus EMS delineations (EMSR312 Ompong, EMSR478 Rolly). Static susceptibility is the share of population inside LiPAD 5- and 25-year flood hazard polygons. Dynamic forcing uses GPM IMERG 3- and 7-day rainfall and rainfall accumulated downstream along the HydroRIVERS network, so towns near river mouths register upstream rain.

Layer 2: Causal lead-lag outbreak graph (stop outbreaks early)

Province-week incidence is log-transformed and seasonally differenced so shared rainy-season peaks do not create false links. PCMCI+ (Runge, 2020) in Tigramite, with partial-correlation tests and lags up to three weeks, recovers a directed lead-lag graph with bootstrap confidence per edge; lagged cross-correlation is the baseline graph. PageRank ranks sentinel provinces that rise first; Louvain groups provinces that move together for coordinated response. Graphs are learned only inside each training window to avoid leakage.

Forecast model

A pooled XGBoost model with province as a categorical feature predicts 1- to 4-week-ahead incidence from lagged cases, calendar week, lagged rainfall, flood exposure (Layer 1) and lagged cases of PCMCI+ leader provinces (Layer 2), trained with the multi-quantile objective at the 23 levels used by the CDC FluSight hub and explained with SHAP. An identical model without graph features is the ablation. Zero-shot baselines set the bar: Chronos-2 (Ansari et al., 2025) with rainfall and flood covariates, TabPFN-TS, a seasonal-naive forecast and a fixed-adjacency MPNN-LSTM. The final forecast is the median ensemble, with conformal (EnbPI) intervals as a check. Models train on earlier years and are evaluated by rolling-origin re-forecasts, never a random split, scored with the Weighted Interval Score (Bracher et al., 2021), relative WIS and coverage.

Layer 3: Access graph and storm scenarios (strengthen healthcare reach)

Nodes are city and municipality centres and hospitals from the DOH National Health Facility Registry and OpenStreetMap. Edges are OSM road links and the 108 province-to-province RoRo routes in the MARINA inventory (roroph), weighted by travel time and frequency. A multi-source shortest-path search gives each town's travel time to the nearest hospital. For each typhoon, ferry routes inside the IBTrACS gale-radius buffer are removed and road links inside the observed flood polygon are slowed or cut; towns with no remaining path are flagged. A raster method, the Malaria Atlas Project friction surface (Weiss et al., 2020) with flooded cells impassable, runs in parallel following Petricola et al. (2022); the disagreement between methods is reported as uncertainty.

Priority score and product

For a chosen typhoon, each province's priority is P = R x A, where R is the forecast probability that incidence over the next four weeks exceeds the province's 90th percentile and A is the share of population cut off from hospital care, so P reads as the expected share of residents at elevated risk who

cannot reach care. The Streamlit dashboard takes a historical or shifted track and returns a priority map, a risk-versus-access 2x2, a ranked table with predicted cases and intervals, cut-off population, sentinel status and SHAP drivers, and an action tier: monitor sentinel sites; pre-position doxycycline and vector-control supplies; or deploy mobile clinics before roads and ferries close. Scenarios are pre-computed, so results return in under one minute.

Validation

We backtest the forecast on Ompong (2018), Tisoy (2019) and the 2019 season, and the access module on all four typhoons, checking whether high-priority provinces saw higher incidence in the following four weeks. Sensitivity to travel speed, ferry rules and storm buffer is reported as rank stability (Kendall's tau). Three target users complete a 15-minute scenario task and the System Usability Scale.

Tools

Python with pandas, GeoPandas, NetworkX, OSMnx, Tigramite, XGBoost, chronos-forecasting, tabpfn-time-series, MAPIE and SHAP; Google Earth Engine; Streamlit; all code in a public GitHub repository with a one-command rebuild.

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APPENDICES

Appendix A. Figure 1. Bantay-Baha pipeline: three graphs, two models, one ranking

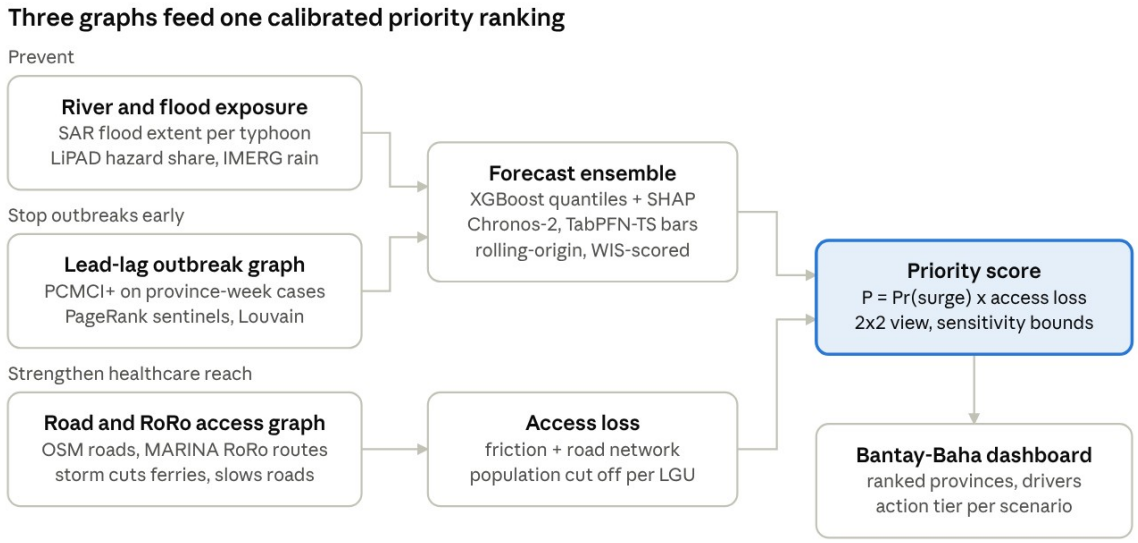


Figure 1. The flood and lead-lag graphs become features of a quantile forecast; the access graph becomes a population-cut-off estimate; the two meet only in the priority score, so each can be validated on its own.

Appendix B. Table 1. Data sources

Data	Source	Role
Dengue cases and deaths, province x week, 2016–Jan 2021	DOH Epidemiology Bureau via Humanitarian Data Exchange (DOH-Epi Dengue Data 2016-2021)	Forecast target; lead-lag graph
National weekly dengue, leptospirosis and other notifiable diseases, 2017–2023	Mendeley Data (Parreño, 2024; v98325dpxv); OpenDengue V1.3	Lepto lag analysis; cross-check
Flood extent per typhoon	Sentinel-1 SAR change detection in Google Earth Engine (UN-SPIDER practice); Copernicus EMS EMSR312, EMSR478; Global Flood Database	Observed flood exposure
Flood hazard maps, 5/25/100-year	UP NOAH / LiPAD (DOST-DREAM)	Static susceptibility
Daily rainfall	GPM IMERG V07 (0.1°); CHIRPS v2.0 (0.05°)	Dynamic forcing
River network with downstream links	HydroRIVERS v1.0 Asia	Upstream rainfall routing
Typhoon tracks	IBTrACS v04r01 (basin WP)	Storm scenarios
Roads and ferry routes	OpenStreetMap (Geofabrik Philippines); MARINA RoRo inventory via roroph (108 routes)	Access graph edges
Hospital locations	DOH National Health Facility Registry; HOT OSM health facilities (HDX)	Access graph destinations
Friction surface and travel time	Malaria Atlas Project 2019 (Weiss et al., 2020)	Raster access method
Population and vulnerability	WorldPop 2020 100 m; Meta High Resolution Population Density (under-5, 60+)	Population weights
Boundaries, PSGC codes, census	HDX COD-AB (PSA/NAMRIA); PSA PSGC 2Q 2026; 2020 CPH	Join keys

Appendix C. Table 2. Development plan, 13–24 October 2026

Dates	Workstream	Deliverable
13–15 Oct	Province-week panel	Parquet panel and data card
13–16 Oct	Flood and climate layers (GEE)	Per-typhoon flood fraction, IMERG features, LiPAD shares
15–20 Oct	Forecasting and lead-lag graph	PCMCI+ graph, quantile XGBoost, foundation-model baselines, WIS table
15–20 Oct	Access network	Friction and network travel times, cut-off population per scenario, sensitivity
19–23 Oct	Dashboard and integration	Streamlit app with pre-computed scenarios
21–24 Oct	Video, deck, repository	3-minute demo, 20 slides, README with one-command rebuild